

Fig. 8: Response time as a function of input overdrive for LM119/219/319

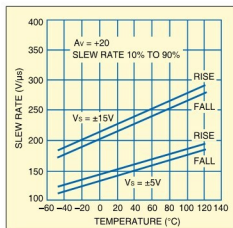


Fig. 9: Slew rate as a function of supply voltage and operating temperature for AD829

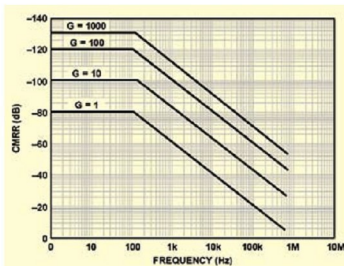


Fig. 10: CMRR as a function of gain and frequency for AD524

the input overdrive changes from 2 mV to 20 mV (Fig. 8).

Dependence of slew rate on both the supply voltage as well as operating temperature is evident from Fig. 9. Slew rate degrades with reduc-

tion in supply voltage and operating temperature. Thus, while using AD829 for an application circuit that is supposed to work over a temperature range of, say, -40°C to $+85^{\circ}\text{C}$, slew rate could vary from $125\text{V}/\mu\text{s}$ to $175\text{V}/\mu\text{s}$ for a supply voltage of $\pm 5\text{V}$.

Similarly, you can notice the strong dependence of CMRR specification of an instrumentation op-amp—one of the most important parameters for this type of op-amp—on both the programmed gain as well as frequency of operation (Fig. 10).

Thermal characteristics

Power dissipation, operating temperature range, maximum junction temperature and thermal resistance (from junction-to-ambient) are the key thermal parameters. As outlined earlier, power dissipation is specified at a certain ambient temperature (usually 25°C) and the specified power dissipation value is such that the junction temperature for the specified value of power dissipation and given thermal resistance value equals the maximum junction temperature. Therefore it must be de-rated for elevated operational temperatures.

Also, the type number designation contains information about the operational temperature range. For example, LM741 and LM741A offer MIL-qualified temperature range of -55°C to $+125^{\circ}\text{C}$, while LM741C offers commercial-grade temperature range of 0°C to $+70^{\circ}\text{C}$. Also, AD829J, AD829A and AD829S offer operating temperature ranges of 0°C to $+70^{\circ}\text{C}$, -40°C to $+125^{\circ}\text{C}$, and -55°C to $+125^{\circ}\text{C}$, respectively.

Other information

In addition to listing the performance parameters and characteristic curves,

comprehensive datasheets include internal schematic to highlight architectural features of the device, package dimensions and application circuits. Some manufacturers even describe the theory of operation of the device with reference to its internal schematic arrangement and explain specific features to highlight the product's capabilities. The application section includes a lot of useful information for the designer. Quite often, the designers get the idea for the intended application circuit from the datasheet.

To sum up

Confining yourself to the information contained on the first page of the datasheet is completely unprofessional when you set yourself out to do an optimal design—one that meets all the design objectives. The impressive look of the first page could more often than not kill your design. The summary of performance specifications and product highlights presented on the first page of the datasheet is intended to introduce you to the product and help you to make a preliminary judgment about the suitability of the device for your application. It should not be treated as the final word.

The datasheet with detailed listing of its parameters, test conditions under which these parameters have been measured, performance curves depicting variation of the parameters with other relevant parameters, recommended test and application circuits, etc reveals the true nature of the device from the perspective of the designer. Therefore a thorough analysis of the datasheet of the device being considered for use should be the designer's imperative and not discretion. ●

Concluded

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